



The mechanistic basis of chromotherapy: Current knowledge and future perspectives



Samina T. Yousuf Azeemi^{a,*}, Hafiz M Rafiq^b, Iram Ismail^{a,1}, Syeda Rabab Kazmi^{a,1}, Ameena Azeemi^c

^a Physics Department, Government Post Graduate College (W) Gulberg, Lahore, Pakistan

^b Physics Department, Punjab University, Lahore, Pakistan

^c CMH Medical College, Lahore, Pakistan

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ABSTRACT

Chromotherapy is a method of treatment that uses wavelengths in the visible region for curing different diseases and medical conditions. Recent advances in photobiology and the speciality of Photobiomodulation are uncovering the cellular and molecular effects of visible range electromagnetic radiation. We discuss the reported effects of visible range radiation on cells (*in vitro* and *in vivo*) and the attempted explanations of the underlying processes with regard to therapeutic effects. Some of the important advances in this area are reviewed, especially the effects of visible light on bacteria, enzymes and the use of visible light for wound healing and treatment of psychiatric diseases for the purpose of explaining the therapeutic implications of chromotherapy. We highlight the correlation of wavelengths used between recently uncovered mechanisms of photobiology and conventional chromotherapy. The elucidation of mechanisms of the cellular and molecular interaction of light will help in deciphering the scientific background of chromotherapy and will help in the application of this alternative therapeutic treatment to many other diseases.

1. Introduction

Chromotherapy (Common names: Colour Therapy, Visible Range Radiation Therapy) is a method of treatment that uses wavelengths in the visible region for curing different diseases and medical conditions.^{1–3} It is one of the oldest therapeutic systems and has been used by the ancient civilizations of India, Egypt and China for the treatment of a number of diseases that include psoriasis, rickets and skin cancer.⁴ Chromotherapy is closely related to light therapy/ photo therapy and Photobiomodulation Therapy/ Low Level Laser Therapy (LLLT). Phototherapy uses polychromatic light and its modern beginnings can be traced to the introduction of an artificial radiation source for healing by a Danish scientist, Niels Ryberg Finsen, who was awarded the Nobel Prize in Medicine for his work on treatment of diseases including lupus vulgaris, tuberculosis and small pox using wavelengths in the visible region.⁵ Low Level Laser Therapy (LLLT)/ Photobiomodulation utilizes near red and infra-red light for healing and regeneration of tissues^{6,7}. Chromotherapy differs from these two therapies in that it strictly utilizes wavelengths in the visible region, namely colours, and hence this methodology is termed as chromotherapy.

Treatment modalities utilizing wavelengths in the visible region have been proven to produce biological effects in molecules, living cells and tissues. Wavelengths in the visible region have been demonstrated to be an effective therapy in a number of medical conditions including Dengue Fever,⁸ Insomnia,⁹ Diabetes,¹⁰ Psychiatric Illnesses,¹¹ Hypertension,¹² Seasonal Affective Disorder (SAD),¹³ Immunity,¹⁴ Hyperacidity,¹⁵ Cutaneous wound healing,¹⁶ Chronic joint disorders¹⁷ and Inflammation.¹⁸ Chromotherapy, Phototherapy and LLLT have been used as complementary and alternative therapies whose mechanisms of action on biological samples are increasingly being understood quantitatively. In the past decade, there has been a renewed interest in the study of cellular and molecular interaction of visible range and near-infrared (NIR) electromagnetic radiation. Photobiomodulation, the mechanistic basis of such interaction, has been explained mostly in terms of light-mitochondria interaction¹⁹ involving the primary photo acceptor cytochrome c oxidase and the associated effects on biological samples. The dependence of a number of diseases on mitochondrial functions has implied that light-induced changes to mitochondria may be the underlying mechanism of the therapeutic effects of light.²⁰ Monochromatic and narrow band light (600–750 nm)

* Corresponding author at: 204 – B Tariq Gardens, Lahore, Pakistan.

E-mail address: saminatazayyen@yahoo.com (S.T.Y. Azeemi).

¹ These authors contributed equally.

has been used for the non-thermal treatment of biological targets. Some of the discovered beneficial effects of light therapy include stimulatory effects on DNA/RNA synthesis and on neuronal growth,²¹ cutaneous wound healing,¹⁶ treating chronic joint disorders¹⁷ and reduction in inflammation.¹⁸ The explanation of mechanisms of such therapeutic effects of visible light has been deemed important due to the increasing use of visible range radiation in diseases and medical conditions and its use in clinical practice for more than 40 years.²²

Light-induced biological effects have been found in both *in-vitro* and *in-vivo* studies. Monochromatic light has shown bactericidal, bacteriostatic and proliferative effects on bacteria in *in-vitro* studies.^{23,24} These particular wavelengths have remained of special interest due to their important role in the elimination of bacteria from infected wounds. A majority of *in-vivo* studies have used direct irradiation of monochromatic light as the primary therapeutic method. One of the important factors in such studies is the light penetration through the cutaneous and epithelial tissue, i.e., direct irradiation is subject to the optical window constraint. Whether visible light can effectively reach the intended cells by the same fluency as used in *in-vitro* studies is an important question in this regard. However, the recent use of subcutaneous light emitting probes and intravascular laser therapy may eliminate this concern completely. The latter methodology, first introduced by Garret et al,²⁵ is carried out by *in-vivo* irradiation of blood by a low power laser (in the range 1–3 mW). The illumination is done through a fibre optic inserted in a blood vessel. The hypothesis behind this therapy is that the circulatory system is responsible for the distribution of the therapeutic effect of visible light on blood, which is characterized by the effect on blood lipids, platelets and the immune system.

Conventional chromotherapy has been administered through various sources such as a monochromatic light, broadband light and recently, a new methodology termed as hydrochromotherapy has shown encouraging results.^{1,3,26,27} Hydrochromotherapy involves irradiation of water samples with monochromatic light, and then intake of that chromotized water. This is an interesting and unique methodology for the administration of chromotherapy. The phenomena behind the preparation of chromotized water has been described as charge quantization, involving formation of ‘hydration spheres’, which were hypothesized after the spectroscopy of chromotized water.²⁶

The mechanisms of these chromotherapy methodologies are being uncovered through recent advances in photobiology, photochemistry and phototherapy. In this review, we will discuss the reported cellular and molecular effects of visible range radiation, which include effects on bacteria – proliferative and inhibitory, effects on enzymes, effects on wounds, molecular effects of hydrochromotherapy and the attempted explanations of the underlying processes. The determination of the cellular and molecular interaction of visible range radiation will be the key to deciphering the therapeutic effect of radiation in the visible region and will help in the application of chromotherapy to many other diseases.

2. Effects of light on bacteria

The effects of visible light on bacteria, both proliferative and inhibitory, have been well discussed in the literature. Such effects have

been considered important due to the bacterial cause of many diseases. Lipovsky et al²³ attempted to identify the most suitable high intensity visible range radiation for inducing a bactericidal effect on *Escherichia Coli* (*E. coli*) and *Staphylococcus aureus* (*S. aureus*) by measuring Reactive Oxygen Species (ROS) production using Electron paramagnetic resonance spin trapping technique. An irradiance of 100 mW/cm² intensity was used for irradiation of cultures. The results indicated the production of more ROS and the reduction in colony counts using high Intensity 415 nm wavelength radiation. *E. coli* reduction up to 99.9% was observed after 10 min of illumination; 90% reduction in *S. aureus* required 20 min of illumination. It was concluded that 415 nm wavelength radiation can be used for bacterial eradication. In this study, the implications of the results on wound healing were also discussed. Since bacteria are commonly found in infected wounds, light therapy, specifically blue light, might be used for bacterial eradication from infected wounds.

Azeemi et al²⁴ studied the effects of low intensity visible range electromagnetic radiation on *E. coli* (*in vitro*) for the purpose of finding out the most effective visible range radiation that induces a bactericidal effect on *E. coli*. Analysis of colony counts of the irradiated *E. coli* samples revealed varying effects of six radiations in the 400–750 nm band. A reduction in colony count was observed in the 538 nm irradiated sample and increased proliferation in 644 nm. Morphological Analysis of *E. coli* was also undertaken using Scanning Electron Microscope (SEM) images of the irradiated samples. The analysis was found to be consistent with the colony counts, with the 538 nm showing bactericidal characteristics and 644 nm radiation causing increased proliferation.

Achieving the same inhibitory effects in *in-vivo* as of *in vitro* irradiation of bacteria with light is an important issue for consideration in the application of chromotherapy. This could have very important implications for the treatment of diseases such as Urinary Tract Infection, where bacteria is the primary causative agent. Whether the wavelengths that show bactericidal effect in *in-vitro* conditions cause the same effects *in-vitro*, is a major question in this regard that needs a scientific explanation. However, it is interesting to note that similarity between the wavelengths used in conventional chromotherapy for eliminating bacteria-caused diseases and that shown to have bactericidal effects *in-vitro*.²⁸

Phototoxicity of visible light has also been reported in a number of other studies (Table 1). From different studies, it is interesting to note the intensity and dose-dependent effects of visible range radiation on bacteria. At different intensities, different wavelengths in the visible region were considered best for reducing the bacteria count. The different experimental results could be also owed to different methods for studying the stimulation of proliferative activity by visible range radiation, due to unavailability of a universal method for studying such type of interaction.²⁹ This highlights the need for the determination of an appropriate protocol for studying the effects of light on bacteria and the design of appropriate clinical trials with controlled parameters for studying the *in-vivo* effects as well.

3. Wound healing through visible range radiation

Numerous studies have elucidated the wound healing

Table 1
Bactericidal effect of different visible range radiation.

Study	Bacterial Strain	Wavelength	Dose [J/cm ²]	Reduction in colony count
Lipovsky et al ³⁰	<i>S. aureus</i> 101	White Light (400–800 nm)	180	99.8%
Maclean et al ³¹	<i>S. aureus</i> NCTC 4135	405 ± 5 nm	23.5	99.6% (Log ₁₀ (N/N0) = 2.4)
Feuerstein et al ³²	<i>P. gingivalis</i>	450 nm	94	99%
	<i>F. nucleatum</i>	450 nm	94	99%
Enwemeka et al ³³	Methicillin Resistant <i>S. aureus</i> (US-300 & IS-853)	470 nm	55	90.4%
Guffey et al ³⁴	<i>P. aeruginosa</i>	470 nm	5	95.1%

characteristics of chromotherapy/low level light therapy, both *in vitro* and *in vivo*. Chaves et al.¹⁶ analysed 20 studies pertaining to the use of LED light for wound healing (*in vitro* and *in vivo*) and identified the use of 456–880 nm wavelengths for cutaneous wound healing. The observed biological effects were reduced inflammation, increased fibroblastic proliferation, increased collagen synthesis and angiogenesis induction. The effects can be characterized by the specificity of the methodologies and the irradiation parameters. The most common irradiation parameters identified are wavelength (nm), irradiance (W cm²), pulse structure and coherence.

Red Light is the most common choice for wound healing due to its increased penetration through tissue because of lower scattering and absorption by tissue chromophores (haemoglobin and melanin).^{16,35} The use of different wavelengths at different stages of wound healing have also been found to accelerate the healing process, with green light being used in the initial stages and red light in the later.³⁶ Light therapy is known to affect three phases of wound healing: inflammatory, proliferative and remodelling. The first phase constitutes the migration of immune cells to the wound, the second phase involves an increase in the activity of fibroblasts together with macrophages, whereas the last phase is characterized by collagen deposition. The characteristic response of different stages of wound healing to different wavelengths points to the existence of more complex underlying mechanisms of light interaction with cell tissue.

The major conjectured mechanism of wound healing is that the inhibitory nitric oxide dissociates from the enzyme under the influence of photons, leading to an accelerated rate of electron transport, greater mitochondrial membrane potential and ATP production. Another hypothesis signifies that light-sensitive ion channels open and allow calcium influx. Photon absorption activates numerous signaling pathways via reactive oxygen species, cyclic AMP, NO and Ca²⁺, leading to activation of transcription factors and increased expression of genes related to protein synthesis, cell recruitment and proliferation, anti-inflammatory signals and antioxidant enzymes.³⁵

4. Effect of visible light on cancer cells

The evaluation of the use of Low Level Laser Therapy as a therapeutic treatment for cancer by observing the *in-vitro* effects on cancer cells is being increasingly carried out, and promising results have been obtained. Peidaee et al studied the *in vitro* effects of visible and far infrared wavelengths of 466 nm, 585 nm, 626 nm (visible light) on human breast cancer cell line (MCF7) and Human Epidermal Melanocytes cell line (HEM).³⁷ Both cell lines were cultured in Dulbecco's Modified Eagle Medium (DMEM) and incubated overnight. Three different exposure and post-exposure regimes were applied on each cell line. Heat shield material was set up to avoid any heating effects. The results suggested that far infrared exposures have a more significant impact on MCF7 cells, with a reduction in the cell viability of MCF7. The viability was measured by PrestoBlue assays and the LDH release activity. The exposure of 3600 nm wavelength was found to have maximum cytotoxicity, which is in line with the finding that 3600 nm wavelength is the activation wavelength of proto-oncogene proteins.³⁸ The viability of normal cells was not found to be affected by the exposures.

In another study, Peidaee et al carried out a qualitative analysis of the effects of applied irradiation on cancer and normal cells. The effect of low intensity light exposures on B16F0 mouse melanoma cancer and CHO (a non-cancer control cell line) cells was done.³⁹ Cells were irradiated with blue, yellow and red visible light (466 nm, 585 nm, 626 nm) and LDH assays and light microscopy was done to evaluate the effects. Results revealed that by increasing incubation and irradiation duration, cell viability of cancer cells decreased when compared with the treated/irradiated normal CHO cells.

5. Interaction of visible range radiation with enzymes

Electromagnetic radiation in visible spectral regions has been shown to affect enzyme activity in living systems. Energy is necessary for every living cell, and organisms absorb that energy from the nutrients. At first level, the chemical energy of the nutrients is not reached to the cell, so it has to be converted biochemically in the form which is absorbable by the cell. Mitochondria is responsible for the transfer of energy and organelles play an important role in this process. The energy range which is transferred from the nutrient to the cell in the form of ATP is also influenced by electromagnetic radiation in the visible region and this range of energy is responsible for the metabolic activity in the intake and also in the energy release system.⁴⁰

ATP synthesis has been found to increase with laser and LED (L-LED) irradiation of mitochondria. This effect was first found out in 1984 when ATP synthesis in rat liver mitochondria was shown to increase following irradiation with He-Ne laser. Moreover, proton gradient difference was higher in the irradiated mitochondria than the non-irradiated.⁴¹ In other related studies, mitochondrial irradiation was followed by an increase in phosphate potential, oxygen consumption and activity of electron chain enzymes.⁴²

The primary mechanism of Photobiomodulation, as proposed by Karu,¹⁹ is that photoexcitation leads to changes in enzyme activity of cytochrome c oxidase, the main photo acceptor molecule. Moreover, changes in the activity of many other enzymes have been observed following irradiation with monochromatic light. Azeemi et al.⁴³ studied the effects of different wavelengths of visible range radiation on superoxide dismutase activity. Superoxide dismutase (SOD) is a key antioxidant enzyme present in all aerobically metabolizing cells. The irradiation of enzyme samples with visible range radiation was carried out according to Shamsuddin methodology.¹ Five different wavelengths were used for the irradiation of the samples. A significant response was seen to different colour wavelengths. Red colour wavelength (644 nm) showed the maximum increase in absorbance following lowered activation energy as compared to all other colours (wavelengths) used, implying that free radical elimination may be enhanced and aided by the application of red colour. He-Ne laser has been found to affect the catalysis of catalase and superoxide dismutase.⁴⁴ Effects of visible light on SOD activity has important implications for free radical elimination, as this process may be accelerated by visible range radiation.

A series of experiments conducted by Bolognani confirmed the irradiation-induced reactivation of enzymes including arylsulphatase, lactate dehydrogenase, myosin ATPase, acid phosphatase, creatine kinase and lactate dehydrogenase.^{45,46} Lipase, Glucose Oxidase and cholesterol esterase + cholesterol oxidase have been found to respond differently to different monochromatic lights, with greatest activation observed at 400 nm and 464 nm.⁴⁷ Light-induced *in-vitro* catalysis of enzymes suggests that *in-vivo* changes may be occurring as well when chromotherapy is administered. Since enzymatic activity is the key to the proper function of the human body, *in-vivo* activation of certain enzymes may be playing a greater role in the process of treatment of diseases through chromotherapy.

6. Effect on human psychology

Chromotherapy has also been used extensively in the modern psychiatric treatment and is based on the fact that different wavelengths in the visible region affect neurohormonal pathways, specifically serotonin and melatonin pathways.¹¹ In the treatment of Post-Traumatic-Stress-Disorder (PTSD), phobias and panic disorder, a special application of chromotherapy on ears named Auricular chromotherapy, has shown promising results.⁴⁸

The basis of utilization of colour therapy in psychiatric treatment is the influence of colours on human psychology and physiology in a number of ways (Table 2) including behaviour, mood attention,

Table 2
Impact of colours on different areas of human psychological functioning.

Area	Aim	Patient Numbers	Colours	Outcomes	Refs
Colors and Cognitive Task Performance	To test the effect of red and blue colour backgrounds while completing a task.	125	Red, Blue	Red colour background more beneficial while completing a simple task; Blue colour background more beneficial while completing difficult task.	50
Colors and Time Perception	To test the effect of red and blue colour on perceived time perception	100	Red, Blue	Perceived time durations are shorter while viewing red colour in comparison to blue colour.	51
Colors and Emotions	To test the effect of blue, green and red colour (with varying hue, saturation and brightness) on arousal and valence.	62	Red, Blue and Green, with three levels for saturation and brightness.	Saturated and bright colours lead to higher arousal. In controlled saturation and brightness, red was most arousing colour.	52
Colors and Memory	To test the effect of colour on memory	48	Red, Yellow. Green and Blue	Colour type influences the memory for the colour of an object. Red and Yellow stick to memory better than blue and green.	53
Colors and Pain Perception	To test whether colours have an impact on pain perception	30	Red, Green, Orange, Blue, Pink, and Yellow	Colors influence pain perception. Red colour given before pain stimuli is perceived as more painful than green and blue.	54

alertness and circadian rhythms.⁴⁹ These effects of color wavelengths are primarily due to the visual effects of the light that falls on the retina, passes through the optic track and reaches suprachiasmatic nucleus (SCN) present in the anterior hypothalamus. SCN has the primary role of maintaining the biological clock, sleep patterns and the circadian cycle. The receptors for melatonin (MT1, MT2) are also present in SCN. SCN takes the information from retina and passes it to the pineal gland which leads to the secretion of melatonin. Serotonin production and secretion increases during the daytime and melatonin production at night time. Many brain disorders such as depressive episodes, PTSD and bipolar disorder are due to low levels of serotonin in the brain. On the other hand, high levels of serotonin lead to hallucinogenic states.¹¹ The maintenance of biological clock in humans is also due to the balance in the production of serotonin and melatonin in the body. Hence, many of psychiatric diseases can be traced directly to the visual effects of visible range wavelengths on human body.

7. Hydrochromotherapy and the role of water in photobiomodulation

The key underlying mechanism of the effect of visible light of biological samples has been described through Photobiomodulation. Photobiomodulation is characterized by the photoexcitation of the photoacceptor cytochrome c oxidase and the associated effects of this light-mitochondria interaction. Recently, an emerging idea of water being the photoacceptor has gained much interest,⁵⁵⁻⁵⁶ and is being described as a ‘Quantum Leap’ in photobiomodulation.⁵⁶ Researches into the intracellular structured water or the exclusion zone (EZ) water have hypothesized the mechanism of photobiomodulation of water. This EZ water, which is formed at the interface of hyrophilic surfaces, has been described as being an energy reservoir: using light to increase its potential energy and acting as a selective supplier of energy for cellular processes.⁵⁷ This supplied cell energy in turn can effect signaling pathways in the cells and can lead them towards or away from programmed cell death.⁵⁶ This phenomena of photobiomodulation of water has also been observed in the irradiation of cancer cells. In a study where red light (670 nm) was used for irradiation of cancer cells, it was found that the irradiation resulted in a change in the viscosity of water layers leading to an increased rate of diffusion across the membrane. This led to the conclusion that multidrug resistance in cancer cells could be overcome through the irradiation, with photobiomodulation being the underlying mechanism.⁵⁸

In conventional chromotherapy, a methodology termed as ‘hydrochromotherapy’ is used,¹⁻³. It involves the intake of ‘chromotized’ water: water that has been irradiated through different wavelengths in the visible region. This water is believed to contain medicinal properties and is used for the treatment of various diseases. After the spectroscopy of this water, it was hypothesized that the irradiation of water leads to the formation of nano structures in water or ‘hydration spheres’ that affect the migration rate of ions within the cells,²⁶⁻²⁷ bearing resemblance to the mechanism proposed in the photobiomodulation of water. This research needs further validation so that the underlying mechanism could be fully understood. In conventional chromotherapy, red light irradiated water is prescribed for use in cancerous conditions and for the mitigation of the side-effects.

8. Implications for chromotherapy

The interaction of visible range electromagnetic radiation with cells has important implications for treatment of various diseases and medical conditions through chromotherapy. Cellular interaction of light has been explained mostly in terms of mitochondrial-light interaction. Since changes to mitochondrial function is the key to the underlying diseases, light induced mitochondrial effects may be one of the major reasons for the therapeutic effect of visible range radiation in various diseases. Red wavelength has been used for curing brain disorders including anxiety,

Table 3

Wavelengths for curing different diseases and the underlying mechanisms of the therapeutic effect of chromotherapy.

Disease/Medical Condition	Wavelength	Underlying Mechanism
Cutaneous Leishmaniasis ^{36,63}	644 nm	Inhibitory effect on Leishmania Tropica
Cholesterol ⁴⁷	400 nm	Changes in the activity of cholesterol esterase
Urinary Tract Infection ²⁸	538 nm	Bactericidal effect on <i>E. coli</i>
Wound Healing ¹⁶	638nm–750 nm	Increased cell proliferation

with the recovery being dependent on changes occurring in cellular mitochondria and enhanced cerebral blood flow.^{6,59} The mitochondrial dependent process of neurodegeneration has been studied in the light of low level laser therapy: an improvement has been found in Alzheimer and depression patients after exposure to LED.⁶⁰ Light therapy has been thought to be beneficial for traumatic brain injury patients, with the stimulation of neurogenesis and angiogenesis being the underlying reason.⁶¹ In this regard, successful results have been obtained on mouse models with 665 nm wavelength showing promising results.⁶² In another study, auricular chromotherapy was found to successfully treat psychological trauma.⁴⁸ Such studies have elucidated the mechanisms of the effects of visible radiation on the human body (Table 3), which in turn has helped us to better understand chromotherapy as a novel alternative therapy in many other diseases.

Light induced in-vitro changes to enzyme activity is a strong factor in the effect of light on biological samples. In a previous study, it was suggested that in-vivo changes to enzyme activity might be occurring as well, and may explain that the same specific wavelengths which were found to increase the enzyme activity, are already being used in conventional chromotherapy⁴⁷.

9. Conclusions

Chromotherapy is a method of treatment that utilizes the visible spectrum of electromagnetic radiation for curing different diseases. Although chromotherapy is a centuries old therapeutic modality, recent advances in photobiology and photochemistry, particularly the cellular and molecular effects of light, are deciphering the scientific background of this alternative therapy. Increasing evidence of micro and macro light induced effects especially on mitochondria, enzymes, bacteria, wounds and the discovery of photo-acceptors provide us with the underlying mechanisms of chromotherapy, which help us in understanding *in-vitro* and *in-vivo* effects of chromotherapy, and are invaluable in determining the application of chromotherapy to many other diseases. Further research in the domain of hydrochromotherapy and the in-vivo effects of monochromatic light needs to be carried out, which will help uncover the novel effects of chromotherapy.

Declaration of Competing Interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

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